A logical qubit with the Steane code in Qiskit

Track 4, Problem 7 · Qiskit Fall Fest Hackathon 2026 · Amritash Mishra (solo) · Qiskit 2.6.0, Aer 0.17.2, Python 3.12

1. Problem

A physical qubit is destroyed by any stray bit flip (X), phase flip (Z) or both (Y). The Steane [[7,1,3]] code stores one *logical* qubit in seven physical qubits so that **any single-qubit Pauli error is detected and undone**. We build the full pipeline (encoder, syndrome extraction, lookup decoder, transversal gates) and ask three measurable questions: (i) are all 21 single-qubit errors (X, Y, Z on 7 qubits) corrected exactly? (ii) what happens with two errors, and at what physical error rate p does encoding start to beat not encoding (break-even)? (iii) does the advantage survive when the encoding and syndrome circuits are themselves noisy? A successful result is a 21/21 table, a logical-vs-physical error curve with the break-even point marked, and an honest answer to (iii).

2. Method

Code. Qubit j (0..6) is labelled by the binary number j+1, giving the Hamming parity-check matrix H = [0001111; 0110011; 1010101]. The six stabiliser generators are X on the support of row r of H and Z on the support of row r, r = 1..3, i.e. X (or Z) on qubits {3,4,5,6}, {1,2,5,6}, {0,2,4,6}. Logical operators are $X_L = X^{\otimes 7}$, $Z_L = Z^{\otimes 7}$. Z-type generators detect X errors and X-type generators detect Z errors; reading the three measurement bits as a binary number gives **error position + 1**, which is the lookup table (a Y error trips both sets).

Encoder (11 CNOT + 3 H, figure below). CNOT 2→4, 2→5 turns a|0>+b|1> into a|000>+b|111> on {2,4,5}, a weight-3 Hamming codeword (an X_L). H on pivot qubits 0, 1, 3 and CNOTs from each pivot onto its stabiliser's other qubits apply $(1+g_X)$ for the three X generators, building $|0_L\rangle$ as the equal superposition of the 8 even-weight Hamming codewords. We verify numerically that all six stabilisers equal +1 on $|0_L\rangle$, $|1_L\rangle$.

Syndrome extraction + feed-forward. Six ancillas, one per generator: Z-type ancilla = CNOTs data→ancilla; X-type ancilla = H, CNOTs ancilla→data, H; all measured mid-circuit into 3-bit registers. Corrections run in the circuit with `if_test((register, v))` → X or Z on qubit v–1. **Transversal gates** are one gate per physical qubit; we extract their 2×2 / 4×4 logical action as $V^\dagger U^{\otimes n} V$ with V the encoder isometry, plus the leakage out of the code space. **Exact error analysis:** for each of the $4^7 = 16384$ Pauli errors we apply the decoder; the residual is harmless iff its X-part and Z-part both have even weight (otherwise a logical X and/or Z was applied), and sum the probabilities with $(p/3)^w(1-p)^{7-w}$. **Noisy circuits:** Aer stabiliser simulator, depolarising noise of total Pauli probability p after every H/CNOT/idle, readout flip p, 7 + 6R qubits for R syndrome rounds. *Implemented ourselves:* everything above. *From libraries:* Qiskit circuits/Aer simulation, numpy/scipy. *Shortcut:* in the circuit-level sweeps the per-round corrections are applied in classical post-processing (a Pauli frame), equivalent to feed-forward for Pauli corrections; true `if_test` feed-forward is used in the noiseless table.

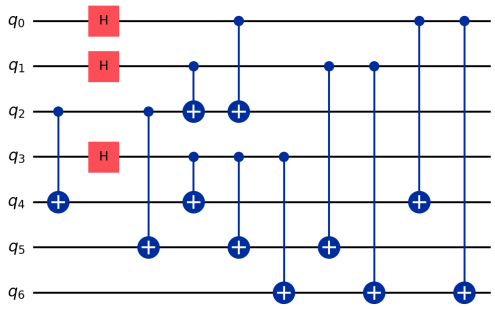


Fig. 1: the encoder (input state on q2).

3. Results

All 21 single-qubit errors are corrected. For each error, and for five test states (|0>, |1>, |+>, |+i>, one random), the circuit encodes, injects the error, extracts the syndrome, corrects by feed-forward, and we compare the data-qubit state with the ideal codeword: minimum fidelity $1 - 4\times10^{-16}$ for all 21 errors and for the no-error control (**21/21 pass**); the 21 syndromes are pairwise distinct (X_j: (0, j+1), Z_j: (j+1, 0), Y_j: (j+1, j+1)). Full table: `figures/single_error_table.png/.csv`.

Transversal gates (leakage < 10^{-14} each): $X^{\otimes 7}$, $Z^{\otimes 7}$, $H^{\otimes 7}$ act as logical X, Z, H; $S^{\otimes 7}$ acts as logical **S†** (diag(1, –i)); $\mathrm{CNOT}^{\otimes 7}$ between two blocks is exactly the logical CNOT. Applying it to $|+_L\rangle|0_L\rangle$ gives a logical Bell pair: all 12 block stabilisers = +1 and $\langle X_L X_L\rangle = \langle Z_L Z_L\rangle = +1$.

Two errors. Single errors never fail (0/21). At weight 2, **147 of 189** Pauli errors cause a logical error: the decoder 'corrects' toward a wrong third position and leaves a weight-3 codeword, which is a logical operator. The 42 survivors are exactly X_iZ_j (i≠j), since X and Z are decoded independently. Circuit demonstration (state θ=1.1, ϕ=0.4): X_0X_1 fidelity 0.67, Z_2Z_5 0.21, Y_0Y_1 0.12, but X_0Z_1 1.00.

injected	X-stab. syndrome	Z-stab. syndrome	correction applied	min fidelity (5 states)	
none	0 (000)	0 (000)	-	1.000000000000	PASS
X0	0 (000)	1 (001)	X0	1.000000000000	PASS
Y0	1 (001)	1 (001)	X0 Z0	1.000000000000	PASS
Z0	1 (001)	0 (000)	Z0	1.000000000000	PASS
X1	0 (000)	2 (010)	X1	1.000000000000	PASS
Y1	2 (010)	2 (010)	X1 Z1	1.000000000000	PASS
Z1	2 (010)	0 (000)	Z1	1.000000000000	PASS
X2	0 (000)	3 (011)	X2	1.000000000000	PASS
Y2	3 (011)	3 (011)	X2 Z2	1.000000000000	PASS
Z2	3 (011)	0 (000)	Z2	1.000000000000	PASS
X3	0 (000)	4 (100)	X3	1.000000000000	PASS
Y3	4 (100)	4 (100)	X3 Z3	1.000000000000	PASS
Z3	4 (100)	0 (000)	Z3	1.000000000000	PASS
X4	0 (000)	5 (101)	X4	1.000000000000	PASS
Y4	5 (101)	5 (101)	X4 Z4	1.000000000000	PASS
Z4	5 (101)	0 (000)	Z4	1.000000000000	PASS
X5	0 (000)	6 (110)	X5	1.000000000000	PASS
Y5	6 (110)	6 (110)	X5 Z5	1.000000000000	PASS
Z5	6 (110)	0 (000)	Z5	1.000000000000	PASS
X6	0 (000)	7 (111)	X6	1.000000000000	PASS
Y6	7 (111)	7 (111)	X6 Z6	1.000000000000	PASS
Z6	7 (111)	0 (000)	Z6	1.000000000000	PASS

Fig. 2: all 21 single-qubit errors: syndrome read from the circuit, correction applied, minimum fidelity over five test states.

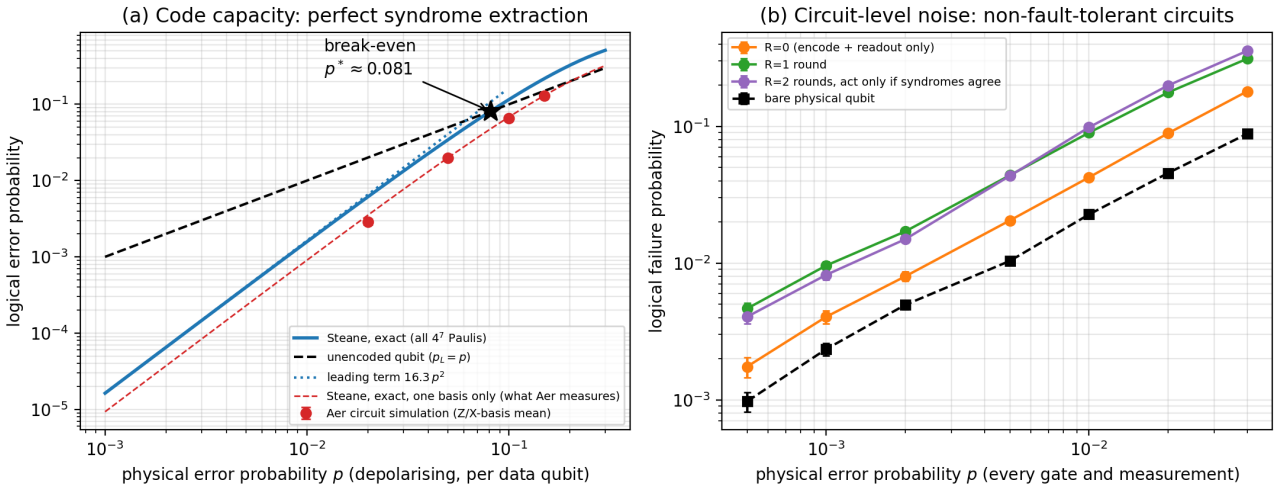


Fig. 3: (a) exact logical error probability vs p (depolarising, perfect syndrome extraction): $p_L \approx 16.3\ p^2$ at small p (= 147/9, the failing weight-2 errors times $(p/3)^2$), **break-even at p* = 0.0811**, above which encoding hurts. Red points: Aer circuit simulation (20,000 shots per basis, error bars = binomial standard error); agreement within ~2σ, and a 10^6-shot circuit-free check at p=0.1 gives 0.0682±0.0003 vs exact 0.0681. (b) Circuit-level noise: encoded vs bare qubit (10,000 shots per basis).

Circuit-level noise: no benefit. At p = 0.001 a bare qubit fails with probability 0.0024; the encoded qubit fails with 0.0040 (encode+readout only), 0.0096 (one syndrome round) and 0.0082 (two rounds, act only when consecutive syndromes agree); at p = 0.005 the numbers are 0.0104 vs 0.0205, 0.0439, 0.0435. The encoded curves are parallel to the bare line (slope 1, not 2) and above it at every p tested (5×10^{-4} to 4×10^{-2}). Reason: each stabiliser is measured with a single ancilla touched by 3–4 CNOTs, so one faulty gate (probability ~ p) can put an error on several data qubits at once, which the distance-3 code cannot undo; the encoder is likewise unprotected. Adding rounds adds more faulty locations, and the 'agree' rule only partly offsets that.

4. Honest limits

• **No fault tolerance:** no flag qubits, no verified ancillas, no Steane-style extraction. The circuit-level result above is what one expects without them, and we make *no* claim that the logical qubit beats a physical one under circuit noise.

• **Break-even scope:** p* = 0.081 holds only for the code-capacity model (noise on data qubits only, perfect extraction). The bare qubit is compared through 'probability of any Pauli error = p'; one-basis and any-logical-error rates differ and are labelled in Fig. 3.

• **Noise model:** depolarising Pauli noise only (no amplitude damping, crosstalk, leakage or idle noise on ancillas); ancillas are fresh, so reset noise is not modelled.

• **Shortcuts:** circuit-level corrections are applied by classical post-processing (Pauli frame); the readout decoder also Hamming-decodes the final data readout. True mid-circuit feed-forward is only exercised in the noiseless table.

• **Statistics:** error bars are binomial only; the stabiliser simulator is exact for these Clifford + Pauli circuits.

• **Not done:** no hardware run (no IBM Quantum access, so no bonus), no logical gates under noise, no non-transversal gate (T needs magic states), no larger codes. None of this is a quantum-advantage claim; a classical Hamming code corrects bit flips only, which is the honest classical comparison for the X-error half of the problem.

5. AI-use note

Claude (Anthropic) wrote the code, tests and this write-up; every result is reproduced by `pytest` and `python src/main.py`. **One thing the AI got wrong:** it passed p directly to Aer's `depolarizing_error(p, n)`, whose parameter is a mixing weight, so the real Pauli probability was 3p/4 (1 qubit) and 15p/16 (2 qubits). **How it was caught:** the Aer circuit simulation disagreed with our exact enumeration (0.0406 vs 0.0681 at p=0.1, >25σ); an effective p of 0.075 reproduced the gap, and after rescaling the two agree. Details in `AI_USE_NOTE.md`.

References

A. M. Steane, Phys. Rev. Lett. 77, 793 (1996); Nielsen & Chuang, *Quantum Computation and Quantum Information*, ch. 10; Qiskit 2.6.0 and Qiskit Aer 0.17.2 (software).